

# MIND-RAG: Multimodal Context-Aware and Intent-Aware Retrieval-Augmented Generation for Educational Publications

Jiayang Yu, Yuxi Xie, Guixuan Zhang\*, Jie Liu, Zhi Zeng, Ying Huang, Shuwu Zhang  
Beijing University of Posts and Telecommunications, Beijing, China

{jiayang-yu, xieyuxi17, guixuan.zhang, ailj, zhi.zeng, ying.huang, shuwu.zhang}@bupt.edu.cn

## Abstract

Although multimodal Retrieval-Augmented Generation (RAG) systems have demonstrated wide applicability, they still suffer from limited image interpretability and limited retrieval performance when processing domain-specific documents. To address these challenges, we propose **Multimodal Intent-Driven Retrieval-Augmented Generation (MIND-RAG)**, a novel framework tailored for educational scientific journals. MIND-RAG introduces two core innovations: (1) **Context-Aware Image Summarization**, which extracts relevant textual context surrounding each image and uses it as a prompt to generate semantic summaries via a multimodal large model, enabling subsequent text-only retrieval; (2) **Multimodal Intent-Aware Reranking**, which jointly infers users' intent based on their latent needs for multimodal querying (e.g. picture, table, text) and educational domain categories, and refines the ranking of retrieved results by aligning document thematic and modality-specific relevance with the inferred query's intent. Evaluated on the MEED-QA benchmark comprising educational journal entries spanning 10 years, MIND-RAG achieves 84.29% accuracy on complex Question Answering (QA) tasks and a 93.5% Mean Reciprocal Rank (MRR) for multimodal retrieval. These results demonstrate the effectiveness of MIND-RAG in real-world scenarios.

## 1. Introduction

Retrieval-Augmented Generation (RAG) systems have recently become a powerful paradigm for enriching large language models with externally retrieved knowledge. In multimodal settings, these systems are extended to leverage not only textual but also figures and tabular information. Despite their promise, existing multimodal RAG systems face significant limitations when applied to domain-specific corpora such as educational scientific publications[6, 11]. These documents often contain dense domain-specific vo-

cabulary and are richly structured with figures and tables, posing challenges for both image understanding and effective retrieval.

Two key challenges emerge in this context. First, *limited image interpretability* impairs the model's ability to ground answers in visual content. Figures in educational publications often contain complex semantic structures that cannot be adequately captured by off-the-shelf image captioning or vision encoders. Second, *limited retrieval performance* stems from the system's inability to fully model user intent, particularly in queries that implicitly demand multimodal information or domain-specific knowledge. As a result, retrieval steps often suffer from semantic drift or fail to return the most relevant data for answer generation.

Several recent studies have attempted to address these issues. For example, Typed-RAG [7] proposes a type-aware multi-aspect decomposition framework that explicitly models the nature of queries and retrieved content, enabling more effective non-factoid question answering across modalities. Complementarily, Mei *et al.* [10] introduce a Multimodal Search Planning module in their MRAG3.0 framework, which detects whether a query requires multimodal retrieval through retrieval classification and query reformulation. While these works represent meaningful progress, most existing approaches still treat modality signals as static inputs and do not dynamically infer users' preferences for specific information types—such as textual descriptions, figures, or tables—which are especially salient in educational contexts. Furthermore, domain specificity (e.g. categorization into subjects like physics or biology) is often overlooked, reducing the applicability of these methods to structured academic corpora. Recent surveys have highlighted the growing landscape of Multimodal Retrieval-Augmented Generation (MRAG) systems and their core challenges[10]. Representative methods such as LightRAG[4], MiniRAG[3], and REPLUG[12] explore different aspects of efficient retrieval, modality fusion, and information routing, which our method complements through intent-aware control.

To address these limitations, we propose **MIND-RAG**

\*Corresponding author.

(Multimodal INTent-Driven Retrieval-Augmented Generation), a novel framework tailored for the retrieval and reasoning challenges posed by educational scientific journals. MIND-RAG introduces two complementary innovations:

- **Context-Aware Image Summarization:** Rather than processing figures in isolation, we extract the surrounding textual context from the document and use it as a prompt for a multimodal large language model (MLLM) to generate semantically rich image summaries. These summaries convert visual content into text, enabling more effective text-only retrieval pipelines.
- **Multimodal Intent-Aware Reranking:** We leverage a large language model to infer whether multimodal retrieval is needed and to classify the query's topical domain. Retrieved results are then reranked based on category-aligned relevance, which mitigates semantic drift and improves the precision of data selection for answer generation.

We evaluate MIND-RAG on **MEED-QA**, a self-constructed benchmark comprising Chinese educational journal entries published over the past ten years. Our framework achieves **84.29%** accuracy on complex question answering (QA) tasks and a **93.5%** mean reciprocal rank (MRR) in multimodal retrieval. These results demonstrate the framework's effectiveness in understanding domain-specific multimodal content and aligning retrieved knowledge with user intent.

In summary, our contributions are threefold:

- We propose MIND-RAG, a novel multimodal RAG framework that explicitly models and exploits user intent in both modality and domain dimensions.
- We propose a context-aware summarization strategy for image understanding within scientific documents.
- We evaluate the proposed framework on a self-constructed dataset, demonstrating its effectiveness in real-world, domain-specific educational scenarios.

## 2. Methodology

MIND-RAG operates through three core stages: multimodal data processing and database construction, multimodal intent detection, and retrieval post-processing with multimodal intent-aware reranking, as shown in Fig. 1.

First, educational PDFs are parsed to extract both textual content (paragraphs, tables) and figures. Each figure is paired with its surrounding context, and a multimodal large language model (MLLM) is used to generate context-aware summaries. These summaries and their corresponding figures together form text-image pairs. The text components are stored in a text database, while the associated figures are saved in the image database. During retrieval, text queries are issued to the text database, and the system returns the corresponding images based on the aligned

text-image pairs. Additionally, a graph structure is optionally constructed to capture implicit entities and relationships within the text. Recent work by Liu *et al.* (2025) demonstrates how context-aware image processing, in the form of structured image summaries, can be highly effective in multimodal retrieval-augmented generation tasks for structured images [9].

Next, given a query, a LLM determines whether multimodal retrieval is necessary and predicts the relevant domain. This intent guides selective retrieval from the appropriate modality-specific databases.

Finally, retrieved candidates from the text, image, and graph databases are re-ranked using a Multimodal Intent-Aware Reranking module, which integrates the predicted intent and domain information to prioritize the most relevant data. The top-ranked context, combined with the query, is then used to generate the final answer.

### 2.1. Multimodal Data Processing and Database Construction

We begin by parsing the input educational PDFs to separately process textual content and visual figures. Text segments are extracted and passed through a large language model (GPT-4o) using a prompt-based approach to extract structured knowledge graphs. These extracted graphs are stored in a dedicated knowledge graph database. Meanwhile, the raw text segments are embedded using a pre-trained embedding model (BGE-M3[1]) and indexed into a text retrieval database.

**Context-Aware Image Summarization** For each image, we begin by identifying its associated textual context. Specifically, we first determine whether the image is accompanied by an explicit caption. If no caption is present, we extract a predefined window of surrounding paragraphs based on the image's original position in the document (set to one paragraph above and one below in our experiments). If a caption exists, we augment this local context by additionally retrieving paragraphs within the same article that explicitly reference the figure based on its caption. This article-level segmentation prevents misalignment due to similar or repeated captions appearing in different articles within the same document. The resulting set of context paragraphs is then concatenated and used as a prompt to guide a multimodal large language model (Gemini 2.0 Flash) in generating a concise, context-aware summary of the image. The generated summary is subsequently embedded using the same BGE-M3 encoder and stored in a dedicated image representation database for downstream retrieval.

This processing pipeline ensures that both textual and visual information are semantically indexed in a unified embedding space, enabling effective downstream multimodal retrieval and reasoning.

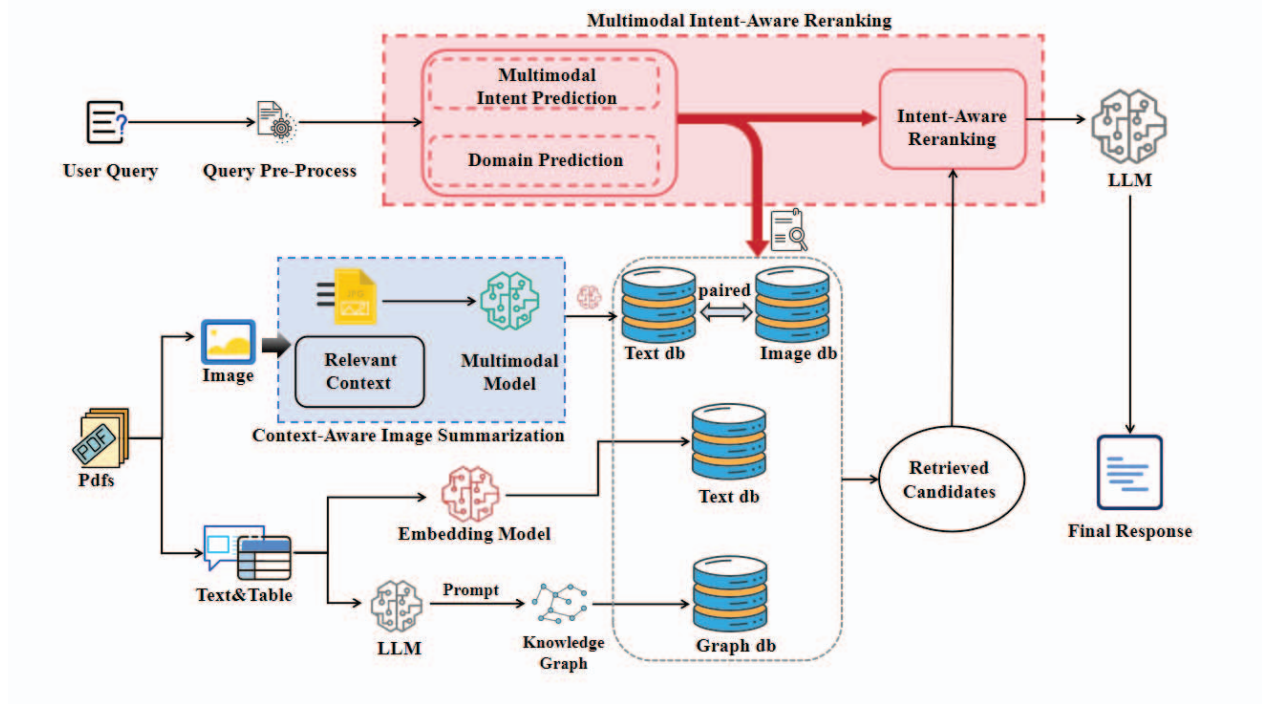


Figure 1. Overall architecture of the proposed MIND-RAG framework, consisting of three stages: multimodal data processing, intent detection, and intent-aware reranking.

## 2.2. Multimodal and Domain Intent Detection

This module determines whether a user query requires multimodal retrieval—text, visual, or both—and predicts its associated domain or topical area. The detected intent signals inform downstream components to select the appropriate modality-specific databases and reranking strategies.

Upon receiving a query, we first apply a lightweight preprocessing stage, which rewrites the input for clarity, extracts relevant keywords, and identifies any timestamp-related information. The timestamp is used later in intent-aware reranking.

We denote the user query as  $q$ . The goal of the intent detection module is to produce a tuple  $(m, d)$ , where  $m$  is the modality intent and  $d$  is the domain label:

$$\mathcal{I}(q) = (m, d) \quad (1)$$

Here, the modality label  $m$  belongs to the set of predefined modality types:

$$m \in \{\text{text}, \text{image}, \text{multimodal}\}$$

In our implementation, the `text` modality includes both unstructured paragraphs and inline tables, as they share similar retrieval pathways and linguistic features. Treating tables as textual elements simplifies modality prediction and ensures consistent downstream handling.

The domain label  $d$  is drawn from a manually constructed domain set  $\mathcal{D}$  based on the characteristics of our dataset. For the MEED-QA dataset, this includes categories such as:

$$\mathcal{D} = \{\text{"Frontier Experiment Design"}, \\ \text{"Equipment Management"}, \\ \dots\}$$

The intent function  $\mathcal{I}(q)$  is decomposed into two separate sub-functions:

$$\mathcal{I}(q) = (\mathcal{I}_{\text{mod}}(q), \mathcal{I}_{\text{domain}}(q)) \quad (2)$$

where  $\mathcal{I}_{\text{mod}}$  predicts the modality intent and  $\mathcal{I}_{\text{domain}}$  classifies the query's domain. Both sub-functions are implemented via prompt-based classification using a large language model.

Two prompt-based intent detection steps are performed using a large language model. The first prediction classifies the query into one of three modality intents: `text-only`, `image-only`, or `multimodal`. This decision governs which retrieval databases will be accessed, as illustrated in Figure 1. The second step assigns a domain label to the query, which is used for domain-aware reranking. Domain categories are predefined based on dataset characteristics.

These intent predictions guide the selection of modality-specific retrieval databases. Retrieved candidates are then

passed to a downstream reranking module for further refinement.

This modular classification approach ensures flexible and interpretable query routing across modalities.

### 2.3. Retrieval Post-processing with Intent-Aware Reranking

The final stage of MIND-RAG refines the retrieved candidate set via a reranking module that incorporates multimodal intent, domain alignment, and semantic relevance. This step is critical for ensuring that the most contextually appropriate data is selected before answer generation.

Given the predicted domain label  $d$ , the top- $K$  retrieved candidates, and their initial similarity scores  $f_1$ , the reranker adjusts each candidate's score based on whether its domain matches the predicted domain of the query. Specifically, as defined in Equation (3), candidates with a matching domain receive a boosted score by multiplying  $f_1$  with a predefined weight  $w$  (set to 1.1 in our experiments), while others retain their original score.

$$S'(q, d_i) = \begin{cases} f_1(q, d_i) \cdot w, & \text{if match} \\ f_1(q, d_i), & \text{otherwise} \end{cases} \quad (3)$$

The re-ranked top- $N$  candidates (with  $N = 5$  in our implementation), along with any modality-specific retrieved content (e.g. images, if required by the predicted intent), are then passed to the large language model, which generates the final answer returned to the user.

Recent work by Chen *et al.* [2] demonstrates that MLLMs can serve as strong rerankers when enhanced with domain knowledge and noise-injected training. Inspired by this, our reranking module further incorporates fine-grained intent signals—both in modality and domain—to dynamically adjust data scores during retrieval. Unlike prior static ranking approaches, our strategy enables context-sensitive prioritization of results based on predicted user preferences and topical alignment.

While the domain-aware reranking improves relevance for textual and image-based retrieval, we also observe unique challenges in the knowledge graph pathway. Specifically, retrieval from the knowledge graph tends to overemphasize surface-level semantic similarity in numeric expressions, particularly for temporal references, due to the fact that entity embeddings are generally not sensitive to subtle numerical differences such as year or version numbers. For example, a query referring to “experiments conducted in 2020” may retrieve entities or procedures from 2021 because the embedding similarity remains high despite the numerical mismatch.

To address this, we introduce a timestamp-aware adjustment mechanism: when a timestamp is explicitly present

in the query, only candidate entities with an exact temporal match retain their original rerank score. Candidates with mismatched timestamps are downweighted by a fixed factor ( $\delta = 0.5$  in our experiments), reducing their influence in the final ranking.

Overall, this reranking stage plays a critical role in our domain-specific setting, where retrieved data often includes technical descriptions with overlapping terminology across domains and modalities. By incorporating domain alignment, temporal precision, and modality-aware reasoning, the reranker helps distinguish semantically similar but contextually irrelevant content.

## 3. Experiments

To systematically evaluate the effectiveness of MIND-RAG in domain-specific multimodal retrieval and generation scenarios, we conduct extensive experiments on a self-constructed benchmark dataset. Our goal is to assess how well the proposed framework addresses the challenges outlined in Section 1, namely (1) limited image interpretability in complex scientific figures, and (2) limited retrieval performance due to the inability to accurately identify users' latent modality preferences and domain-specific intent.

To this end, we organize our evaluation around the following research questions:

- **Visual Understanding and Retrieval Impact:** Can MIND-RAG effectively extract relevant information from complex figures and leverage this information during retrieval to achieve superior performance compared to other methods, thereby addressing the challenge of limited image interpretability in domain-specific documents?
- **Intent-Aware Reranking Effectiveness:** Can MIND-RAG effectively infer users' latent modality preferences and domain-specific intent using large language models, and leverage this inferred intent to guide reranking and improve retrieval precision? Does incorporating these intent signals into the reranking process lead to measurable improvements in actual retrieval results?
- **Overall Effectiveness and Comparative Advantage:** Is the proposed MIND-RAG framework effective as a whole in enhancing retrieval accuracy and answer quality, when compared to existing baseline methods? On which evaluation metrics does our approach demonstrate particularly strong improvements?

### 3.1. Dataset and Setup

We construct a domain-specific multimodal QA dataset, named **MEED-QA** (Modern Educational Equipment Domain QA), based on approximately 10,000 scientific articles published over the past ten years in a national journal focused on educational technology. The corpus consists



of richly structured documents featuring textual content, figure-caption pairs, tables, and domain metadata, offering a realistic and challenging testbed for multimodal retrieval-augmented generation.

To comprehensively evaluate the performance of our framework, we define two distinct testing splits within MEED-QA, each targeting a specific retrieval or reasoning capability. Each test set consists of queries, intermediate retrieval ground-truths, and final answer ground-truths, enabling the computation of quantitative evaluation metrics:

- **Multimodal Single-hop QA:** 400 questions designed to evaluate basic retrieval performance and the effectiveness of context-aware image summarization. These questions can be answered using a single modality-specific item (*e.g.* a paragraph, figure-caption pair, or table).
- **Multimodal Multi-hop QA:** 300 questions that require reasoning over multiple pieces of data spanning different modalities—such as combining textual descriptions, figures, and tables—aimed at testing the system’s capability for compositional inference capabilities.

Evaluation of our model is conducted using a comprehensive suite of metrics, designed to capture both retrieval quality and answer generation fidelity:

- **Answer Relevancy:** Measures the semantic alignment between the predicted and ground-truth answers, serving as a core indicator of overall QA quality.
- **Context Recall@k(R@k):** Quantifies the proportion of gold-standard supporting contexts successfully retrieved within the top-3, top-5, and full retrieved set, respectively—reflecting data coverage.
- **Context Precision@k:** Evaluates the proportion of retrieved contexts that are actually relevant, highlighting retrieval precision under different cutoff thresholds.
- **Rouge-L:** Captures the longest common subsequence between the generated answer and the reference, assessing fluency and content overlap. This metric was introduced by Lin [8] and is widely used in automatic summarization and text generation tasks.
- **Mean Reciprocal Rank (MRR):** Evaluates the ranking quality of the first correct context among retrieval candidates, emphasizing early precision in retrieval. This metric is commonly used in retrieval tasks, such as those in the Open Graph Benchmark [5], to assess multimodal retrieval performance.

### 3.2. Visual Understanding and Retrieval Performance

To assess whether MIND-RAG can effectively retrieve and interpret multimodal data—especially visual content such as complex figures—we focus on the **Multimodal Single-hop QA** subset of MEED-QA. This split consists of 400

questions, each of which can be answered using a single source of information—such as a figure, a textual paragraph, or a table—from the same document.

We compare MIND-RAG against the following baselines:

- **CLIP-based Crossmodal Retrieval:** A contrastive baseline using the CLIP model to encode image inputs without incorporating any associated textual descriptions (*e.g.* captions). During retrieval, the system computes embedding similarity directly between the query and candidate images, relying solely on visual information for matching.
- **RAG (w/o Figure Captions):** A variant of MIND-RAG where the retrieval process does not involve searching for context associated with figure captions. This baseline tests whether the explicit matching of figure captions to relevant context contributes to improved retrieval and answer performance.

For all methods, we adopt `bge-m3` as the sentence embedding model for textual retrieval. When handling multimodal inputs, we utilize `Gemini-2.0-Flash` as the vision-language model to process and summarize images. The extracted figure captions and visual summaries are then incorporated into the retrieval pool as text data.

Table 1. Comparison on Multimodal Single-hop QA subset.

| Method                    | MRR          | R@3          | R@5          | R@all        |
|---------------------------|--------------|--------------|--------------|--------------|
| Clip-based                | 0.358        | 0.304        | 0.391        | 0.478        |
| RAG (w/o Fig. Cap. Match) | 0.848        | 0.826        | 0.870        | 0.870        |
| <b>MIND-RAG (Ours)</b>    | <b>0.935</b> | <b>0.917</b> | <b>0.943</b> | <b>0.975</b> |

We report four retrieval metrics in Table 1: Mean Reciprocal Rank (MRR), Recall@3, Recall@5, and Recall@all.

As shown in Table 1, MIND-RAG outperforms all baselines across all retrieval metrics, achieving an absolute improvement of 8.7% in Recall@3 and 8.7% in Recall@all over the strongest baseline. This demonstrates that intent-aware retrieval and figure-caption grounding play a critical role in enhancing answer accuracy.

The Simple Multimodal baseline, which uses BGE over naive concatenated inputs without explicit figure grounding, performs significantly worse (MRR: 0.358). This confirms that merely aggregating heterogeneous inputs is insufficient for effective multimodal reasoning.

The RAG variant without figure-caption matching achieves competitive performance (MRR: 0.848), but still lags behind MIND-RAG. This gap highlights the importance of aligning visual data with relevant textual context, rather than treating all modalities uniformly.

Overall, these results validate our hypothesis that explicit modeling of user intent and figure-caption grounding leads to more precise retrieval and higher-quality responses in

multimodal QA tasks.

### 3.3. Intent-Aware Reranking Effectiveness

To assess the impact of our proposed reranking mechanism, we examine how well it can enhance the relevance and contextual appropriateness of retrieved results when guided by inferred modality preference and domain alignment. Specifically, we compare the reranked results from MIND-RAG against standard retrieval approaches that do not utilize modality-aware reranking.

We conduct a series of experiments using both the single-hop and multi-hop subsets of the MEED-QA dataset. These subsets are designed to test different levels of reasoning complexity: the single-hop subset includes questions that can be answered with a single piece of evidence, while the multi-hop subset requires integrating multiple sources across documents.

We compare MIND-RAG’s reranking approach against several baseline methods, including:

- **No Rerank:** The initial retrieval is performed without any reranking mechanism.
- **Traditional Rerank Retrieval:** Retrieval candidates are ranked using a general-purpose reranker (bge-reranker-v2-m3) based solely on similarity between the query and the retrieved text, without incorporating modality-specific signals or domain intent

Table 2. Comparison of reranking methods on Multimodal Single-hop QA

| Metric                | No Rerank | Traditional Rerank | MIND-RAG      |
|-----------------------|-----------|--------------------|---------------|
| Answer Relevancy      | 0.8157    | 0.8385             | <b>0.8672</b> |
| Context Recall@3      | 0.8753    | 0.8942             | <b>0.9170</b> |
| Context Recall@5      | 0.8965    | 0.9182             | <b>0.9435</b> |
| Context Recall@all    | 0.9421    | 0.9567             | <b>0.9753</b> |
| Context Precision@3   | 0.7238    | 0.7594             | <b>0.7791</b> |
| Context Precision@5   | 0.6875    | 0.7123             | <b>0.7410</b> |
| Context Precision@all | 0.6105    | 0.6419             | <b>0.6627</b> |
| Rouge-L               | 0.4172    | 0.4354             | <b>0.4978</b> |

Table 3. Comparison of reranking methods on Multimodal Multi-hop QA

| Metric                | No Rerank | Traditional Rerank | MIND-RAG      |
|-----------------------|-----------|--------------------|---------------|
| Answer Relevancy      | 0.7892    | 0.8115             | <b>0.8429</b> |
| Context Recall@3      | 0.8683    | 0.8876             | <b>0.9085</b> |
| Context Recall@5      | 0.9105    | 0.9317             | <b>0.9562</b> |
| Context Recall@all    | 0.9394    | 0.9521             | <b>0.9712</b> |
| Context Precision@3   | 0.7026    | 0.7465             | <b>0.7823</b> |
| Context Precision@5   | 0.6582    | 0.6931             | <b>0.7268</b> |
| Context Precision@all | 0.5873    | 0.6187             | <b>0.6534</b> |
| Rouge-L               | 0.4072    | 0.4283             | <b>0.4526</b> |

As shown in Table 2 and Table 3, MIND-RAG consistently outperforms both traditional reranking and the no-reranking baseline across all evaluated metrics. The improvement is especially prominent in answer relevancy and

context recall, with MIND-RAG achieving 0.8672 answer relevancy compared to 0.8385 for traditional rerank and 0.8157 for the baseline. This suggests that incorporating modality and domain signals allows the model to better surface contextually appropriate results.

In terms of context recall and precision, MIND-RAG exhibits a strong balance between coverage and selectivity. For example, it achieves 0.9753 in overall context recall and 0.6627 in overall context precision, indicating that it can retrieve more relevant data while minimizing noise. These improvements contribute to more informed and grounded generation.

Taken together, these results demonstrate the effectiveness of intent-aware reranking in enhancing both retrieval quality and downstream answer generation, in multimodal single-hop QA and multimodal multi-hop QA settings.

### 3.4. Overall Performance and Comparative Advantage

To evaluate the overall effectiveness of the proposed MIND-RAG framework, we conduct a comparative study against two representative baselines: (1) a conventional Retrieval-Augmented Generation (RAG) system that does not incorporate any knowledge graph or intent modeling, and (2) a graph-enhanced RAG system that utilizes traditional entity linking and graph retrieval mechanisms but lacks dynamic modality-aware reranking.

All models are evaluated on the MEED-QA benchmark under the same experimental setup. We focus on key evaluation metrics including Answer Relevancy, Context Recall, Context Precision, Faithfulness, Rouge-L, and a composite Weighted Score. The results on the single-hop QA subset are summarized in Table 4. The results on the multi-hop QA subset are summarized in Table 5.

Table 4. Comparison of MIND-RAG with Graph-RAG and Basic RAG baselines on single-hop QA.

| Metric                | Basic RAG | Graph-RAG | MIND-RAG      |
|-----------------------|-----------|-----------|---------------|
| Answer Relevancy      | 0.8261    | 0.8280    | <b>0.8672</b> |
| Context Recall@3      | 0.8687    | 0.8964    | <b>0.9170</b> |
| Context Recall@5      | 0.8792    | 0.9201    | <b>0.9435</b> |
| Context Recall@all    | 0.9485    | 0.9465    | <b>0.9753</b> |
| Context Precision@3   | 0.7572    | 0.7580    | <b>0.7791</b> |
| Context Precision@5   | 0.7021    | 0.7106    | <b>0.7410</b> |
| Context Precision@all | 0.6321    | 0.6402    | <b>0.6627</b> |
| Rouge-L               | 0.4450    | 0.4670    | <b>0.4978</b> |

Overall, the quantitative results clearly highlight the comparative advantage of MIND-RAG across both single-hop and multi-hop QA tasks. Notably, MIND-RAG consistently outperforms the Graph-RAG and Basic RAG baselines in nearly all evaluation metrics.

For instance, in the single-hop setting, MIND-RAG achieves an Answer Relevancy of 0.8672, outperforming Graph-RAG (0.8280) and Basic RAG (0.8261). It also

Table 5. Comparison of MIND-RAG with Graph-RAG and Basic RAG baselines on multi-hop QA.

| Metric                | Basic RAG | Graph-RAG | MIND-RAG      |
|-----------------------|-----------|-----------|---------------|
| Answer Relevancy      | 0.7810    | 0.8002    | <b>0.8429</b> |
| Context Recall@3      | 0.8698    | 0.8924    | <b>0.9085</b> |
| Context Recall@5      | 0.8810    | 0.9312    | <b>0.9562</b> |
| Context Recall@all    | 0.9506    | 0.9541    | <b>0.9712</b> |
| Context Precision@3   | 0.6917    | 0.7153    | <b>0.7823</b> |
| Context Precision@5   | 0.6520    | 0.6695    | <b>0.7268</b> |
| Context Precision@all | 0.5713    | 0.5951    | <b>0.6534</b> |
| Rouge-L               | 0.4236    | 0.4360    | <b>0.4526</b> |

exhibits higher Context Recall across different retrieval depths—achieving 0.9170 at Recall@3 and 0.9753 at Recall@all—demonstrating its ability to surface the correct supporting data more reliably. Corresponding improvements are also observed in Context Precision and Rouge-L, indicating more concise retrieval and higher answer fidelity.

In the multi-hop setting, MIND-RAG similarly leads with an Answer Relevancy of 0.8429, alongside Recall@5 of 0.9562 and Precision@5 of 0.7268. These consistent gains suggest that MIND-RAG is effective at handling both simple and compositional questions through intent-aware retrieval and modality-sensitive reranking. The Rouge-L score of 0.4526 further supports its capacity to generate coherent and faithful responses.

These findings collectively validate the effectiveness of integrating multi-intent representation, modality-awareness, and reranking mechanisms into retrieval-augmented generation.

## 4. Conclusion

We present MIND-RAG, a multimodal and intent-aware retrieval-augmented generation framework specifically designed for educational scientific publications. MIND-RAG addresses two key limitations of existing multimodal RAG systems: poor visual interpretability and inadequate modeling of user intent. Through context-aware image summarization and multimodal intent-aware reranking, our system improves both evidence retrieval precision and downstream answer quality.

Evaluated on the MEED-QA benchmark, which includes both single-hop and multi-hop educational QA tasks, MIND-RAG achieves state-of-the-art performance across all evaluated metrics. The framework demonstrates strong generalization in both text-only and multimodal settings, effectively aligning retrieved content with users’ modality preferences and domain-specific information needs.

Our ablation studies further confirm the importance of each proposed component, showing that both context-aware visual grounding and intent-driven reranking contribute substantially to system performance. These findings under-

score the importance of dynamically modeling both modality and domain intent in educational retrieval scenarios.

Future work may extend this approach to broader document types and multilingual settings, as well as integrate more sophisticated graph-based reasoning for even deeper semantic understanding.

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